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'We are the oceans, we are the people!'[1]: fisher people's struggles for blue justice

Irmak Ertör

ABSTRACT
Fisher people and their communities around the world have been key social actors in seafood production and they are still feeding the world with about two-thirds of the catches destined for direct human consumption. However, they are subjected to a wide range of injustices due to coastal and offshore investment projects and the inequalities embedded in the global capitalist marine economy. These injustices are manifested through socio-environmental conflicts, which have been studied and mapped in the Environmental Justice Atlas (EJAtlas) together with other social actors' resistances. This article examines 120 fisher folk conflicts that have been identified, documented and mapped in the EJAtlas, while proposing a Blue Justice framework to understand their underlying causes and how fisher people confront such injustices. As a grounded theoretical approach emerging from the scrutiny of fisher resistances and building on environmental justice, critical marine sociology and political economy literatures, the Blue Justice framework proposed here focuses on three dimensions: (i) *material and biophysical dimension*, (ii) *spatial justice,* and (iii) *autonomy and sovereignty*. In adopting this framework, this analysis reveals the blue layers of environmental justice by uncovering both the causes of conflicts faced by fisher communities and their political agency.

KEYWORDS
Environmental justice; blue justice; marine social metabolism; ocean grabbing; fisher conflicts; fisher struggles

Fisher people as actors of blue justice

Small-scale fisher communities around the world are facing significant material, ecological, social, and political challenges (Barbesgaard 2018; Bavinck, Jentoft, and Scholtens 2018). The primary challenge is linked to the industrialization and intensification of capture fisheries from the 1950s onwards, which by the 1980s had resulted in several collapses of fish stocks, the extinction of a range of species – especially of predators but including small pelagic fish – and a circular but continuous fall in global yields (and biomass) with declining global primary production in seas and oceans (Pauly et al. 1998; Pauly et al. 2003; Chassot et al. 2010). Reconstructed catch models incorporating

[1]This was the slogan of the World Forum of Fisher Peoples' (WFFP) 7th General Assembly, which took place in New Delhi, India, in November 2017. See its theme song by the Media Collective: https://www.youtube.com/watch?v=SY-hKP-5Azw. See also the blog text by Masifundise on the 7th General Assembly (8 December 2017): https://www.masifundise.org/oceans-waters-people-7th-general-assembly-world-forum-fisher-peoples-asserts-small-scale-fishers-human-rights-food-sovereignty/

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an estimation of illegal, unreported or unregulated (IUU) fisheries – but excluding by-catch, which is estimated to be about 30 percent of the entire global landings – suggest that global fisheries landings have been shrinking by around 500,000 metric tons per year since the late 1980s (Pauly et al. 2003). Many fisheries that had been considered renewable but also – by way of the fisheries industry – as 'inexhaustible' resources have shown signs of reaching or overpassing their ecological limits (ibid.).

This intensification in industrial capture fisheries, combined with the concentration of capital in the hands of a few big fishing companies, has led to the deepening of inequalities between industrial and small-scale fishers. As discussed by Campling (2012), distant-water fishing fleets operating in open seas with more abundant fishing sources have expanded to new marine frontiers and have also entered traditional fishing areas of small-scale fisher people, which has created further injustices and conflicts between the two groups. Clark, Longo, and Auerbach (2019, 196) frame this process as an 'unequal economic and ecological exchange' created by the global capitalist marine economy and seafood production system.

A comparison between large-scale and small-scale fishing in regards to their ecological and socio-economic dynamics illustrates this unevenness. First, the number of fishers employed in small-scale fisheries constitutes 'more than 90 percent' of employment in the sector, excluding aquaculture (FAO 2020, 133). Moreover, catch per ton of fuel consumed is estimated at approximately 1 to 2 tons in the industrial fishing fleet – and annual fuel consumption amounts to 37 million tons – whereas it is about 4 to 8 tons in small-scale fishing – with annual fuel consumption around 5 million tons (Pauly 2006). Discarded fish and other seafood levels approach 8 to 20 million tons in industrial captures, while discards are quite low in small-scale fishing (ibid.). Additionally, it is important to note that even though they have been considered 'backwards' during this industrialization process (Ertör-Akyazi 2020), small-scale fisher people are estimated to produce about two-thirds of the catches destined for direct human consumption but still live under increasingly vulnerable and marginalized conditions (FAO 2015).

Faced with serious shrinkage in both local and global fish stocks, the industrial seafood sector has transitioned and invested in intensive (marine) aquaculture to provide further fish supplies as well as to expand marine commodity frontiers (Saguin 2016; Ertör and Ortega-Cerdà 2019). As a 'false solution' to declining fish captures (Lee and Cloutier de Repentigny 2018) or as a 'protein fix' (Brent, Barbesgaard, and Pedersen 2020), intensive industrial aquaculture remains greatly questionable regarding its claims of sustainability (for a discourse analysis unpacking such claims, see Ertör and Ortega-Cerdà 2017). Intensive industrial aquaculture in many places has rather displaced fisher people from their traditional territories and entrenched a range of socio-environmental conflicts and environmental injustices (Adduci 2009; Bogadóttir 2020).

However, these are only the challenges that fisher people face in relation to the injustices surrounding seafood production. There exists a much broader political and economic agenda on, in, and around the seas, often subsumed under the term Blue Growth, that currently threatens fisher people more acutely than in previous decades (Barbesgaard 2018; Bennett et al. 2021). Blue Growth strategies aim to ensure further investment in marine areas through the intensification and expansion of already existing marine sectors – such as coastal and marine tourism and (intensive) marine aquaculture – and the promotion of new ones like seabed mining, marine biotechnology, and ocean

energy. Most of these sectors are highly industrialized and their rapid expansion has already shown signs of their socially and ecologically devastating potential (Ertör and Hadjimichael 2020). Many of these industrial 'extractivist' (Acosta 2013) projects have proven to be harmful at various levels for small-scale fisher people and communities in different geographic contexts. With the Blue Growth agenda globally gaining support from powerful industries and policy-makers (Ertör and Hadjimichael 2020), together with the expansion of marine commodity frontiers into new and marginal(ized) areas (Campling 2012; Saguin 2016), these transformations will not only constitute ecological problems but also create socially and politically fraught conflicts, wherein I assert that the main social actors are small-scale fisher people resisting continuous growth and destructive expansion.

In the global context of falling fish stocks, combined with other negative anthropogenic effects such as climate change, pollution – caused by a range of investment projects and urbanization placed on the coasts – and rising inequalities regarding access to and control over marine commons, small-scale fisher communities have organized and mobilized themselves as social actors to confront the grabbing of their aquatic / marine resources and the accompanying process of environmental and 'fisheries injustices' (Mills 2018). In order to understand their resistance, this article focuses on the socio-environmental conflicts of fisher communities around the world, who are facing various challenges posed by a range of coastal and off-shore investment projects adversely affecting them. Building on the literature on environmental justice (Walker 2009; Martinez-Alier et al. 2010, 2016; Schlosberg 2013; Agyeman et al. 2016; Scheidel et al. 2020) and fisheries justice (Mills 2018), and following the methodology of the Environmental Justice Atlas (EJAtlas) – explained in detail below – I will examine and demonstrate the roots of these conflicts by using a political-ecology lens and framing Blue Justice as an expanding subfield of environmental justice.

Accordingly, I will first discuss the context of the broader political-economic context leading to inequalities and injustices faced by fisher folks, followed by a review of the existing literature on marine and blue justice (and marine sociology). Additionally, this article will propose the framework of 'Blue Justice' as a new thematic and conceptual area in the environmental justice debates and movements (for a detailed discussion of the environmental justice literature, see Agyeman et al. (2016), and for its application in marine issues, see Martin et al. 2019). Second, I will show that these socio-environmental conflicts (also called ecological distribution conflicts by Martinez-Alier et al. 2010; Scheidel et al. 2018, 2020) are a result of both increasing (marine) social metabolism and escalating attempts of grabbing marine resources for further capital accumulation. Thus, by scrutinizing 120 'aquaculture and fisheries conflicts' identified and documented in the Environmental Justice Atlas (EJAtlas),[2] I aim to give a global overview of these conflicts and fisher struggles. Fisher communities and mobilized groups have not been sufficiently visible in conflicts attracting global public attention, but often rather perceived as incidental actors supporting some of the resistance movements demanding environmental justice. In this article, I intend to show that fisher people are mobilized social actors demanding fisheries and environmental justice, building alliances and stopping some detrimental projects thanks to their local, regional, and global resistances.

[2]See the online database at: www.ejatlas.org

The article is structured as follows: Section 2 provides a political economic overview of global fisher struggles and touches upon the issues of justice and social class in fisheries. Section 3 scrutinizes the environmental justice literature and situates 'Blue Justice' as one of its theoretical and practical expansion areas by proposing a three-dimensional theoretical framework. Section 4 highlights the methodology, mainly based on the EJAtlas with complementary qualitative research methods. Section 5 analyzes global fisher conflicts with 120 cases and discusses them via the Blue Justice framework. The final section concludes the article.

Political economic context of global fisheries conflicts and the question of justice

Before addressing the Blue Justice framework proposed in this article, a closer look at the distinctive characteristics of fisher conflicts, as well as fisher organizations, is necessary to discuss social class and the structural power relations in which they operate. Marine social science literature proposing blue justice, marine justice, and environmental justice frameworks – scrutinized and built on in the next section – rarely touch upon broader political economic discussions and the social class dimension of the mobilized people that make justice claims. As Bavinck, Jentoft, and Scholtens (2018) noted, there is only limited critical marine social science research that also considers, criticizes, and situates fisher struggles, structural power relations, and collective social action.

Key studies with this intention have focused on class, collective action and movement formation in the context of fisheries. One such study comes from Sinha (2012) (in a crucial collection advancing critical political economy studies on fisheries, edited by Campling, Havice, and McCall Howard 2012), who explored the class dimension in the Indian context of fisheries transformation and the formation of national and transnational fishworker movements. Other studies include Scholtens (2017, 934), who discussed collective action in the context of fisheries as a 'typically contestatory strategy of ordinary people to pursue their claims for social justice' and Bavinck, Jentoft, and Scholtens (2018), who conceptually framed social struggles in fisheries contexts. From a global perspective, however, the marine social science literature lacks detailed class analysis in fisheries, a concept that has been discussed in much further detail in research focusing on peasants.

To explore the class dimension and formation of (justice) movements in the context of fisheries, one must first understand the global industrialization, capitalization and intensification of fisheries as thoroughly discussed by Campling, Havice, and McCall Howard (2012). This transformation, which accelerated especially after the 1950s, reshaped the class dimensions in fishing and led to a highly capitalist class in fisheries dominated by a few industrial fishers, while slowly pushing others out of this new model or exploiting them as wage laborers (Sinha 2012). As Sinha (2012) explains in the Indian context, this process included the emergence of the capitalist class in fisheries and the evolution of the national and transnational fishers movement as a reaction and alternative incorporating small-scale fisher people and organizations from a wide range of social classes with different levels of access to the means of production. In this context, small-scale fishers, who have very diverse material, ecological, cultural, political, and class characteristics in

each region, have been continuously marginalized as oppressed social classes resisting the new neoliberal seafood production regime.

Discussing current environmental and labor abuses and injustices in seafood production system, Clark, Longo, and Auerbach (2019, 196–197) claim that:

> The capitalist system, as the social organization of generalized commodity production, has established distinct patterns of expropriation of marine species and fishery workers, which is in part built upon the unequal economic and ecological exchange between the global South and North. These dynamics have influenced fishing operations at sea, and the processing of fish on land, contributed to overfishing, led to the development of environmentally damaging aquaculture operations in Southeast Asia, and fostered global social inequality and environmental injustice.

As such, the authors uncover 'how the global division of labor associated with marine commodity production creates specific patterns of exploitation, degradation, and unequal exchange' (p. 197). Often, these workers abused in industrial fishing boats are migrants who previously have been small-scale fishers in their hometown (TNI 2020).

A clear distinction between mobilized small-scale fisher organizations from around the world is that some primarily practice subsistence fishing – usually combined with other rural work like small-scale agriculture – while others engage in small-scale and/or artisanal commercial fishing. In most countries, the official definition of small-scale fishing emphasizes boat length, i.e. usually smaller than 12 or 15 meters, and/or fishing gear, that is less mechanized fleets with less ecologically destructive fishing equipment (Chuenpagdee et al. 2006). Hence, one can observe that this has led to very distinctive social classes and material, cultural, and political relations, not only between industrial and small-scale fisher (SSF) groups but also among small-scale fishers themselves. However, this article will show that those in the fisher movements still coalesce around a range of common justice claims – connected issues around justice, autonomy and alliance building which will be discussed in the next section.

Therefore, while small-scale fisher organizations are very diverse and not homogenous, coming from different social classes in terms of access to the means of production – mainly boats, fishing gears and initial capital – in this article, I build on the literature on transnational agrarian (Edelman and Borras 2021) and fisheries (Sinha 2012) movements forming new collective identities of resistance against 'a common enemy' (McMichael 2006). Such resistances need to involve 'blending an anti-capitalist, or class, politics with a campesino [peasant and/or fisher] politics that revalues substantive forms of economy (…) via a distinctive modern politics rooted in a global moral economy' (McMichael 2006, 415). This article thus follows the lenses offered by Martínez-Torres and Rosset (2010, 150):

> In the neoliberal era, supranational corporations and institutions dictating neoliberal policies have negatively affected most sectors of society. One of the consequences of this is that class or cultural differences are no longer the barrier they once were for transnational collective action. In fact, rural organizations and peasantries around the world share the same global problems even though they confront different local and national realities.

While they talk about the formation of La Vía Campesina (the Peasant Way) as a transnational social movement, most of the fisher organizations I will discuss in this article form

part of the global fisheries movements, such as the World Forum of Fisher Peoples (WFFP) and the World Forum of Fish Workers and Fish Harvesters (WFF).

As argued by Martínez-Torres and Rosset (2010, 152), both the neoliberal agribusiness and seafood production sectors led to a food system that reshaped state-society-capital relations, in which new forms of social movements emerged that are 'more autonomous, horizontal, and more based on collective identities rather than just social class'. Therefore, as Gallar-Hernández (2021, 6–7) discussed regarding the Peasant Action Schools in Spain, to create 'collective political subjects' as part of a broader struggle is an exercise in 'how subaltern social groups organize and build their own mechanisms of political and cultural [resistance and] strengthening grapple[s] with the ideological and material pressure exerted by dominant classes', namely capitalist industrial fishers in this case.

In this article, I will touch upon this debate by uncovering global justice movements of small-scale fisher communities, mostly against their capitalist and industrialized counterparts as well as against the attempts by other coastal industries grabbing their resources. My aim is thus uncovering their ways of collective action formation – as an oppressed/subaltern class – and alliance building as well as establishing a close link between environmental justice and fisher folks' struggles while situating their resistances in the broader political economic context of capitalism (and ocean and resource grabbing attempts discussed below). This discussion will enable to uncover the political economic roots of fisher struggles and understand that these are not isolated local conflicts but rather some form of collective action.

Blue layers of environmental justice

Political ecology and ecological economics literatures focusing on environmental justice have incorporated an extensive and detailed study of socio-environmental conflicts – also called ecological distribution conflicts – around the world (Martinez-Alier et al. 2010). Outstanding and diverse analyses from this perspective include Gerber et al.'s work (2009) on tree plantation conflicts, Özkaynak and Rodriguez-Labajos' study (2017) on mining conflicts, Avila's research (2018) on wind energy conflicts, and studies from a global perspective on socio-environmental conflicts and environmental justice movements (Martinez-Alier et al. 2016; Scheidel et al. 2020). However, the conflicts in which fisher folks have been the primary mobilized social actors have remained rather understudied in the literature on environmental justice (for a few exceptions, see Veuthey and Gerber 2012; Ertör and Ortega-Cerdà 2015).

As a new and timely thematic evolution, 'Blue Justice' needs to take its place in the expanding field of environmental justice. Recently, there have been significant attempts by a range of scholars and fisher rights advocates to establish blue justice and marine justice frameworks. Especially worth highlighting are the efforts of the Too Big to Ignore (TBTI) network[3] and their publications focusing on different geographic regions,[4] as well as other projects such as the Blue Justice Initiative. In addition, Bennett et al. (2021) framed blue justice as a response to blue growth initiatives, while

[3]TBTI defines itself as a 'global research network and knowledge mobilization partnership' and its name refers to the slogan that 'Small-Scale Fisheries are Too Big To Ignore'. For more information see: toobigtoignore.net

[4]See Jentoft (2019) for his book and chapter on Blue Justice; Isaac's effort (2019) to frame fisheries debates from a human rights approach, TBTI Communication Brief on Blue Justice (2019) as well as a forthcoming book in TBTI Global Book

touching upon environmental justice. Finally, Widener (2018, 414) framed marine justice 'as a paradigm and potentially a movement' – similar to others such as the environmental and climate justice movements – while Martin et al. (2019, 234) built on Widener's work and applied environmental justice lenses in 'five conceptual domains – space, time, knowledge, participation in decision-making, and enforcement – in which attention to marine environments resonates with and expands environmental justice framings'. These have been the first explicit attempts to bring together environmental justice debates and marine social sciences scholarship.

In this article, I aim to develop this framework on the intersection of environmental and blue justice further. I claim that the Blue Justice framework (both in terms of its theory and practice) needs to: (i) go beyond simple policy suggestions within the existing capitalist marine economy and politics and incorporate the recent discussions of environmental justice and 'critical environmental justice' (CEJ) – emphasizing intersectional, multi-scalar, anti-authoritarian and indispensable aspects (Pellow 2018) – as well as (ii) shed light on the root causes of socio-environmental conflicts faced by fisher and coastal people from a broader political economic perspective. This scrutiny and synergy are necessary to understand fisher struggles better as well as to establish their links with resistances taking place in other conflictive industrial projects, where a range of actors such as peasants, students, worker unions, and local communities struggle for environmental justice. While environmental justice theories and movements have moved from 'reactive protests against environmental injustices to advocacy for sustainable and just alternatives to address inequalities' (Agyeman et al. 2016, 325), recent research has begun to uncover different social actors' socio-environmental justice claims such as 'fisheries justice' (Mills 2018) and their links with other justice movements – such as climate, food, and energy justice – (Agyeman et al. 2016; Mills 2018) and especially food sovereignty movements (Levkoe, Lowitt, and Nelson 2017).

As noted by Agyeman et al. (2016, 330) in their article exploring new trends and directions in environmental justice practice and scholarship, there is a need for broader frameworks which tie together a range of contemporary social movements and justice struggles, 'as our environment is increasingly disrupted by a highly neoliberal and capitalized world marked by increasing inequality and vulnerability'. Blue Justice framework, as represented in this article, therefore aims to open up a new theoretical, empirical, and thematic space for environmental justice practice and scholarship. This framework will provide a methodological and conceptual tool to better understand the struggles of fisher communities and the injustices and inequalities they face, as well as to challenge these power relations and inequalities.

By establishing this link and proposing a theoretical framework for exploring 'Blue Justice', I will focus on three dimensions that allows for understanding the underlying causes of SSF struggles, revealing their political agency, and examining the blue layers of socio-environmental injustices. These three dimensions build on existing debates regarding the evolution of the environmental justice field (especially inspired by the works of Walker 2009; Martinez-Alier et al. 2010, 2016; Agyeman et al. 2016; Pellow 2018; Scheidel et al. 2020) and literature on transnational social movements and food

Series on Blue Justice from an interactive governance perspective. The concept was also discussed extensively in the 2019 MARE Conference.

sovereignty (Martínez-Torres and Rosset 2010; Sinha 2012; Gallar-Hernández 2021), while also emerging from a grounded approach through the close examination of fisher resistances.

Below, I will first explore the *material and biophysical dimension* of these conflicts and the Blue Justice framework. Here I draw on the social metabolism perspective and explore the recent changes in marine social metabolism leading to Blue Justice struggles. Second, I will scrutinize *spatial justice* – as an important dimension of Blue Justice, in which I will predominantly scrutinize displacement and dispossession of fisher people as a result of ocean grabbing. The third aspect of Blue Justice I aim to delve into is the struggle for *autonomy and sovereignty* of fisher communities. All three aspects enable better situation of fisher folks' struggles around the world by exploring different layers of injustices they are faced with. This framework will also help to uncover their alliances with a range of similar justice and sovereignty movements with the common goal to overcome these deep injustices and inequalities.

Material and biophysical dimension of blue justice

In order to understand the material and biophysical dimensions of fisher folk conflicts, I will first focus on the rising (and changing) marine social metabolism. In this article, I utilize the social metabolism concept by drawing on Martinez-Alier et al.'s approach (Martinez-Alier et al. 2010; Scheidel et al. 2018) focusing on energy and material extraction, transformation, exchange, use, and waste generation, as well as by following the eco-Marxist tradition by referring to the 'complex, dynamic interchange between human beings and nature' (Foster 2000, 158; Clausen and Clark 2005, 429). Drawing on Longo and Clark's work (2016), in which they frame marine sociology using social metabolic analysis, I use the concept of 'marine social metabolism' as a theoretical lens to illustrate the human relationship with marine environments and their material impacts throughout the entire input-throughput-output relations. This framework also needs to be situated within the broader political economic processes by 'merging the material basis of human existence together with the social processes associated with the larger-scale historical structures that influence the interpenetration of society and nature' (p. 467). By engaging with these theoretical debates, my first aim is to reveal the material and biophysical dimensions of the Blue Justice framework.

Building on Bavinck, Jentoft, and Scholtens's (2018, 47) argument that 'conflict provides a useful entry point for examining the nature of social struggle in fisheries', I will also adopt the perspective that views socio-environmental conflicts shaped and influenced by different cultural perceptions and valuations of nature, while at the same time arising from the material metabolism of the economy (Martinez-Alier et al. 2010, 127). Hence, by examining the inequalities embedded in these conflicts throughout the entire input-throughput-output relations, I will first shed light on changes in marine social metabolism with respect to recent trends of the capitalist blue economy.

Peak fish as a result of increasing marine social metabolism

Currently, the marine economy is experiencing a sharp increase in the use of energy and materials and circulation(s) in regards to marine spaces and resources (Jouffray et al. 2020). This global rise in material and energy throughput, emerging from specific uses

and allocations of marine and coastal areas, limits the traditional uses of SSF communities and challenges their livelihoods. The first characteristic of rising marine social metabolism affecting SSF communities is the transition to capital- and energy-intensive methods of seafood production, both through industrial capture fisheries and intensive marine aquaculture. The movements of industrial fishing fleets demonstrate the expansion of capture fisheries and the unequal distribution of marine harvests.

There exist many examples of this industrial expansion and unequal harvest and catch distribution. In the EU, industrial fishing fleets obtain more than a quarter of their captures from outside EU waters, often by extending their operations to the Global South through 'Fisheries Partnership Agreements' with Cape Verde, Ivory Coast, Sao Tomé and Principe, Gabon, Madagascar, Senegal, Comoros, Seychelles and Mauritius (Doerr 2016). French and Spanish fleets have established domination over the industrial fishing sector in the Western Indian Ocean since the 1980s (Campling 2012). Arriving from Asia, Chinese industrial fishing fleets have expanded into Senegalese and Malagasy (Madagascar) waters (EJAtlas 2018b), while Turkish trawlers and purse seiners have recently moved to Western Africa, especially to Gambia, Somalia, Mauritania and Senegal (Philippe 2020; Ertör-Akyazi 2020). The arrival of these industrial fishing fleets has often led to increased inequalities and conflicts with local SSF communities (DuBois and Zografos 2012; Ertör-Akyazi 2020).

An essential part of this phenomenon is linked to overfishing, mainly a result of industrial fishing – including some Illegal, Unreported and Unregulated (IUU) – operations. This has brought about the term 'peak fish' to the debates of fisheries governance and the fishing economy (Pauly et al. 2003; Clausen and York 2008; Pauly and Zeller 2016) and is critical to understanding the harsh industrialization and expansion of capture fisheries. The 'peak' concept is commonly understood as the unsustainable consumption of non-renewable fossil fuels and minerals. However, since fish stocks – which are in fact renewable resources – were perceived as a 'non-exhaustible resource' by influential parts of the industrial fishing sector and faced with such a harsh and aggressive expansion and industrialization of global capture fisheries, fisheries reached their 'peak point' already by the end of the 1980s (Pauly et al. 2003). The accompanying rush and expansion of fisheries to the sea for further capital accumulation, along with accompanying anthropogenic industrial activities, has threatened the regeneration and maintenance of sustainable fish stocks. Therefore, focusing on the *material and biophysical dimensions* of Blue Justice, it is important to note that 'the growth potential of living or biotic resources is naturally limited, and therefore, cannot fuel exponential economic [*in this case*, *blue*] growth' (emphasis added, EJOLT Glossary 2012).

Increasing use of energy and materials accompanied by waste and contamination on coasts

In addition to the capitalization and industrialization of seafood production, the expansion and intensification of resource extraction (gas, oil, and minerals) and rising commercial marine transportation have increased the use of energy, waste generated, and materials obtained from marine ecosystems. Likewise, the establishment of thermal and nuclear power plants on coastlines usually leads to increased water temperature and stratified seawater due to thermal discharge in marine and coastal areas (Huang, Lin, and Zheng 2019) as well as to increased pollution resulting from chemical or

nuclear waste. New investments and energy generation projects, both coastal and offshore, are becoming a part of the recent rush to the sea, leading to rising marine social metabolism of our society (Longo and Clark 2016). This rise results in severe adverse impacts on SSF communities, deepening existing inequalities in the access to and control over common marine (and also terrestrial) spaces and resources and often causing conflicts between industrial interests and fisher people. Therefore, the underlying material causes of fisher folk conflicts and resulting impacts are global and structural in nature, due to both the current capitalist food system and continued political neglect and marginalization of these communities.

In a period in which 'Blue Growth' policies and strategies are at the center of political debates (as discussed and criticized by Barbesgaard 2018; Ertör and Hadjimichael 2020) and the seas and oceans are perceived as a new frontier enabling further commodity expansion and capital accumulation (Ertör and Ortega-Cerdà 2019), any attempt to further increase the marine social metabolism might create even deeper injustices and more intense conflicts with fisher folks.

Spatial justice: struggles against ocean grabbing and displacement

The second dimension of the Blue Justice framework emphasizes *spatial justice*. A key work on spatial justice is Walker's analysis of 'multiple spatialities of environmental justice' (2009). Walker suggests that for 'both revealing inequalities and understanding the processes through which these are (re)produced', the multiple spatialities of environmental justice need to be scrutinized (p. 615). Hence, with the same purpose, uncovering the multiple spatialities of Blue Justice is one of the concerns of this research. This dimension is critical in order to establish the link between *material and biophysical* side of fisher folk conflicts and their complex and dynamic 'scaled *spatial relations* and flows' (p. 615), as well as understanding fisher organizations' strive for *autonomy and sovereignty*.

Under the *spatial justice* dimension, I will explore different ways in which the 'grabbing' concept emerged in marine social science scholarship. I argue that ocean grabbing attempts by private and/or state actors create significant forms of spatial injustices through a set of displacement and dispossession processes targeting fisher communities.

'Ocean grabbing' is a term that has its roots in 'land grabbing' discussions promoted by networks of scholar-activists and social movements claiming food justice and food sovereignty (GRAIN 2008; Borras et al. 2011; Borras and Franco 2012; Hall et al. 2015). Building on the term 'land grabbing', 'ocean grabbing' was initially defined as 'capturing of control by powerful economic actors of crucial decision-making around fisheries, including the power to decide how and for what purposes marine resources are used, conserved and managed now and in the future' and as the 'processes and dynamics that are negatively affecting the people and communities whose way of life, cultural identity and livelihoods depend on their involvement in small-scale fishing and closely related activities' (TNI 2014, 3). Following the debates of fisher movements and their allies, the term was also adopted in academic discussions; either to explore the evolution of the concept as done by Bennett, Govan, and Satterfield (2015), to criticize blue growth initiatives (Barbesgaard 2018), or to study the politics of different marine sectors such as seabed mining (Zalik 2015, 2018), fisheries in East Africa (Doerr 2016) or in New Brunswick (Knott and

Neis 2017), and real estate and touristic development and conservation initiatives in the Colombian Caribbean (Pérez 2019), among others.

The link between *spatial justice* and ocean grabbing becomes most prominent here. As discussed by Hall et al. (2015) on land grabbing, ocean grabbing is not limited to the appropriation of physical space but also materializes through ocean 'control grabbing' (Barbesgaard 2019). The accumulation processes achieved via ocean grabbing projects occur through fishers' displacement, dislocation, and dispossession. The inequalities and injustices produced in these processes have both material and spatial dimensions, resulting in further marginalization of fisher peoples (Bavinck et al. 2017). Building on Doerr's study (2016, 1) on fisher folks in East Africa highlighting how 'capitalist core countries try to circulate their own crises of over-fishing and relocate it geographically into the periphery', I aim to track these injustices and inequalities – deepened by further dispossession processes – around the world by uncovering and analyzing the socio-environmental conflicts and struggles of fisher folks against ocean grabbing attempts.

Thus, in this article, I use the ocean grabbing phenomenon to refer to different political-economic efforts led by state and private actors that limit the access of fisher folks to marine spaces and marine commons, take away their control over the space, ways and means of production, and degrade their relations with the seas and oceans. The global movements of fisher people (especially WFFP and WFF) also use terms like 'resource grabbing' or 'aquatic space grabbing', in part to recognize the demands and claims of freshwater fishers. For this article, however, I will focus on the marine aspect, since marine and freshwater seafood production have distinct characteristics, species and methods of production – thus, the process of grabbing also displays differences.

It is important to note that the globally rising marine social metabolism scrutinized in the subsection above is not the unique cause of fisher folks' conflicts. Rather, it occurs in combination with particular political decisions that allow environmental and fisheries injustices and ocean (and control) grabbing by powerful economic and political actors which leads to the conflicts, as discussed below. While increasing marine social metabolism provides the material basis of these conflicts, political and cultural power structures certainly shape material outcomes as well. Therefore, the *spatial* dimension of Blue Justice implies how and why political decisions result in where and how investment projects are placed and which groups are displaced and dispossessed. The spatiality of Blue Justice thus gives us a deeper understanding of the processes of displacement, dispossession, marginalization and even the criminalization of fisher folks, which further limits their political agencies and capabilities for mobilization. A theoretical lens to understand fisher folks' conflicts thus needs to incorporate both the material and spatial dimensions as well as uncover their political agency as local resistances and global social movements, which will be further scrutinized below.

Autonomy and sovereignty: political agency and alliances of fisher folks

The two dimensions of Blue Justice explored above, i.e. the *material and biophysical* and *spatial justice* dimensions, relate to the third dimension, that of *autonomy, self-determination, and sovereignty*. This third dimension enables the analysis of fisher folks' conflicts in terms of their political agencies and the formation of social movements.

Firstly, the dimension of *autonomy and sovereignty* enables the discovery and understanding of the political space local struggles open up through their resistances against industrial investment projects on the ground. Secondly, it clarifies how these local struggles 'scale up' their local mobilizations and establish links and solidarity with other fisher struggles as well as how they build alliances with other social actors.

Third, each movement includes different social groups that are suffering from increased marginalization due to their 'intersectional' minority and 'vulnerabilized' identities (Crenshaw 1989; Malin and Ryder 2018). This is also a challenge presented by Pellow (2018, 39) to EJ scholars, since 'multiple forms of inequality drive and characterize the experience of environmental injustice'. Thus, it is important to explore how gender, class, race, and ethnicity further compound the marginalization of different subaltern groups in the same resistance and how these movements attempt to incorporate the *autonomy* of such groups into their broader struggles. Finally, this dimension points out the political possibilities and potential for alliances between movements struggling for justice – such as fisheries, climate, food justice – and sovereignty – like food and energy sovereignty. This dimension therefore will provide a conceptual lens to discover the links and political potential of different social and environmental justice organizations struggling towards a more radical political transformation of society as a whole.

Using these theoretical lenses and departing from Martinez-Alier et al.'s (2016) argument highlighting the existence of a 'global environmental justice movement', I aim at uncovering the blue layers of the global environmental justice movement through the analysis of fisher folks' mobilizations. In order to understand fisher peoples' shared claims for justice and scrutinize their political agency, collective action formation and alliance-building, it is essential to analyze their conflicts and struggles around the world, which will be discussed below with detailed data on specific conflicts. *Autonomy and sovereignty* are thus important concepts to consider when analyzing justice claims stemming from the identified socio-environmental conflicts. This perspective facilitates the understanding that these socio-environmental conflicts are not isolated local conflicts, but rather a result of the *material* and *spatial* injustices fisher people confront, often embedded historically and politically in the capitalist food system and its geographical, as well as gender, race, and class-based, unevenness. As a result, their local and global struggles show the ways in which they strive for *autonomy* and *sovereignty*.

Methodology

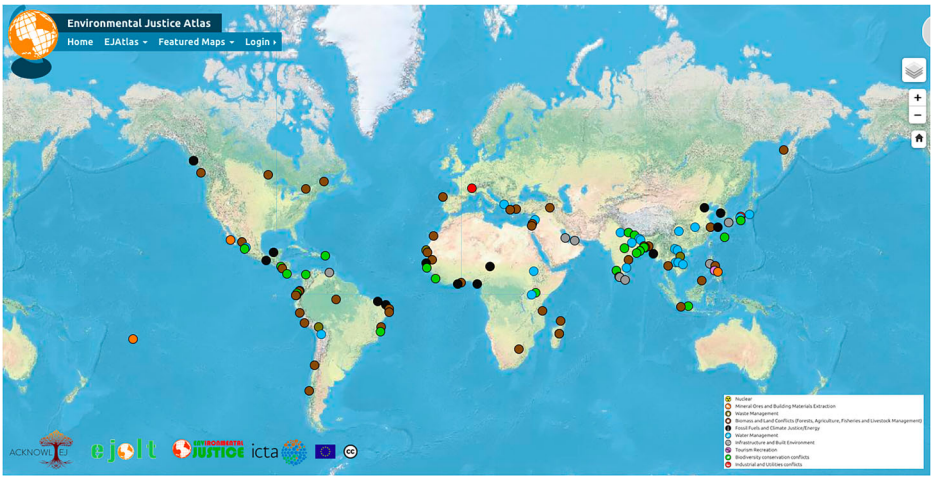
This research primarily relies on the methodology put forth by the Environmental Justice Atlas (EJAtlas), the largest global database on environmental conflicts, whose approach is discussed in detail by researchers forming part of the EnvJustice project[5] (Martinez-Alier et al. 2016; Scheidel et al. 2020). The EJAtlas was first established in 2011 as a global map indicating and analyzing ecological distribution conflicts. In the following years it has continued as a collaborative knowledge production process undertaken by academics and civil society. As of March 2020, the EJAtlas has compiled a structured data set on 3,093 conflict cases from around the world. The EJAtlas' unit of analysis is defined by Scheidel et al. (2020, 4) as 'an environmental conflict provoked by a specific state- or corporate-

[5]For detailed information on the ERC-funded EnvJustice project and its academic outcomes see: www.envjustice.org

driven resource use project (e.g. a hydroelectric dam, or a mine) due to perceived risks and negative socio-environmental impacts triggering mobilizations'. While some of these projects and resulting conflicts are widely known, many usually do not appear in the mainstream media nor result in formal legal action or court cases, through which they could be easily tracked. However, a variety of such (observed or latent) adverse socio-environmental, economic, or health impacts lead to different degrees and types of conflicts and mobilizations by a range of actors. These lesser known conflicts may be identified through a combination of methods including: scrutinizing local (alternative) media sources, field work, studying or participating in local community-based efforts by local actors, organizations, and wider networks struggling for environmental justice. This is how and why the methodology of the EJAtlas increases the visibility of such socio-environmental conflicts and provides a platform for collaboration between such networks.

The documentation of socio-environmental conflicts in the EJAtlas is accomplished by filling a standardized questionnaire of both multiple-choice and open-ended questions. The information gathered includes data on (i) the historical evolution of the conflict case; (ii) its exact location; (iii) the specific resource(s) at stake – usually fish or crustaceans like shrimp in fisheries conflicts; (iv) details of the investment project causing conflict or the grabbing of local people's common resources – as well as the companies, financial institutions and government actors involved; (iv) the socio-economic, health and environmental impacts of the project; (vi) the social actors that are mobilized against this project; (vii) Environmental Justice Organizations (EJOs) and initiatives supporting the (fisher) struggles; (viii) their mobilization strategies – ranging from official complaint letters to occupying public spaces; (ix) and the outcomes of the conflict – e.g. whether it stopped the project, strengthened alliances between (fisher) people and other social actors advocating for environmental justice, or established broader networks, among others. All the provided data is then verified by an academic team from the EnvJustice project and experienced moderators at the EJAtlas, and when necessary, complemented and evidenced with secondary sources such as scientific articles, published reports, newspaper articles, relevant legislations, and/or available audiovisual materials in order to ensure their consistency and compliance with the methodology of the EJAtlas database (for a more detailed discussion on the methodology of the EJAtlas, see Scheidel et al. 2020).

The EJAtlas comprises ten main categories indicating the primary reason(s) leading to the conflict, e.g. a resource use project or an investment project (see the legend of Map 1 showing the ten 1st level categories). These include: biomass and land conflicts – the main category including industrial or illegal fishing in traditional fishing territories and fish or shrimp farming, nuclear energy, mineral ores and building material extraction, waste management, fossil fuels/energy and climate justice conflicts, water management, infrastructure and built environment conflicts, tourism recreation conflicts, biodiversity conservation conflicts, and industrial and utilities conflicts. As mentioned above, among the global and diverse socio-environmental conflicts, the EJAtlas has already registered and documented cases about fisheries conflicts under the main category 'biomass and land conflicts (forests, agriculture, fisheries, and livestock management)' (Martinez-Alier et al. 2016). However, in neither the academic literature nor among the studies relying on the EJAtlas has there been an explicit focus on global fisher folks' conflicts until



Map 1. Aquaculture and fisheries conflicts presented in the EJAtlas (Source: www.ejatlas.org).

now, as this is a rather recent research agenda introduced through the EJAtlas and EnvJustice project.

Building on the methodology of the EJAtlas, this study analyzes the data on fisher folks' conflicts documented in the EJAtlas under the second-level category, 'aquaculture and fisheries conflicts', some of which are also defined as 'ocean and resource grabbing'. Although most of the 'aquaculture and fisheries conflicts' are categorized under the biomass and land conflicts – as it is the default main category – my dataset composed of 120 cases – categorized under the 2nd level 'aquaculture and fisheries' conflicts – also presents diverse 1st level categories such as fossil fuels and climate justice/energy[6] or tourism recreation, to which they are linked (see Supplementary Table 1).

A substantial part of the data regarding SSF conflicts has been compiled by the author herself and in collaboration with her scholar-activist colleagues – from both within and outside the EnvJustice project coordinating the EJAtlas, fisher and food sovereignty movements, local and regional initiatives, and Environmental Justice Organizations (EJOs) working on the ground with fisheries and environmental justice claims.

Additionally, the author has used participant observation and participatory action methods by attending and collaborating in the 7th General Assembly of World Forum of Fisher Peoples (WFFP) held in New Delhi, India, in November 2017; in the WFFP Europe regional meetings in Thessaloniki, Greece, in November 2018; as well as through an ongoing collaboration of more than five years with the Turkish member of WFFP, Istanbul Birlik (the Association of Istanbul Fishing Cooperatives). In-depth interviews with WFFP members from various countries such as Ecuador, Honduras, India, Mauritania, Spain, Mexico, Thailand, and Indonesia, as well as with representatives of other fisher struggles such as the WFF (World Forum of Fish Workers and Fish Harvesters) and NAMA (North American Fisheries Alliance) contributed to data collection. The data

[6]Cases in which oil spills occurred and affected the marine ecosystem and fishing area (see Supplementary Table 1 and next section for concrete instances) as well as proposed gas pipeline construction or offshore hydrocarbon extraction projects, are conflict examples under this category, where the fisher folk entered in conflict with fossil fuel-based industries either due to observed contamination or to the potential threat they pose.

documented in the EJAtlas, gathered through in-depth interviews and participant observation methods and complemented by secondary sources (publications in academic journals, global or regional fisheries reports, videos, press releases, local newspapers and a range of social media accounts of fisher struggles) were analyzed through open-coding methods.

The conflict cases identified, documented, and mapped in the EJAtlas database thus form the foundation of both the empirical material and evidence used to support my hypotheses regarding blue layers of the environmental justice movement as well as global fisher peoples' struggle for fisheries and blue justice. The mapping of global fisher folk conflicts in the EJAtlas contributes to making these struggles more visible and indicates that these are not isolated local cases, but rather a result of global material, political and economic dynamics of expanding commodity frontiers and ocean and resource grabbing – as discussed in further detail below.

Mapping and analyzing global manifestations of fisher struggles through Blue Justice framework

This section presents an analysis of fisher folks' socio-environmental conflicts documented by the EJAtlas using the three dimensions of the proposed Blue Justice framework. Despite multiple studies focusing on local or national fisheries conflicts in the literature, until now there has not been any global analysis of fisher people's conflicts around the world. This article aims to fill this gap and reflects upon 120 conflicts[7] (indicated in the Map 1 and demonstrated in detail with their names, countries, and corresponding categories[8] in the Supplementary Table 1), in which fisher people and their communities are among the main mobilized actors, if not the leading mobilized group. It intends to situate fisher communities' struggles from around the world into the agenda of environmental justice (and blue justice) practice and scholarship through uncovering, visualizing, and analyzing the injustices faced by small-scale fisher people and their communities.

The Blue Justice framework developed above and the use of the EJAtlas to analyze the 120 cases of fisher folks' struggles documented in this article enable the systematic tracking of: (i) the impacts of increasing marine social metabolism, (ii) both local and global ocean grabbing and dispossession attempts as well as the inequalities and injustices faced by fisher people, and (iii) the alliances and collective action formation by fisher people.

Some of the most famous and globally well-known environmental conflicts such as the Chevron oil spill in Ecuador (North 2015), toxic waste dumping in North Carolina (Bullard 2018), ZAD ('Zone à defender', eng. *Zone to Defend*) (Bulle 2020), Blockadia,[9] Standing Rock (EJAtlas 2020), or global Climate Strikes[10] are not conflicts in which fisher people are at the forefront of the resistance. However, fisher people are in fact actively participating and mobilizing to defend their fishing rights and oppose state- or private sector-led

[7]As of 31 March 2020, there have been 120 conflicts documented in the 'Aquaculture and fisheries' second level category from among 3,093 identified and documented cases in the global EJAtlas database (www.ejatlas.org).
[8]Different types of investment projects – displayed with varying colors on the map – leading to conflicts with fisher communities can be seen both in the Map 1 and the Supplementary Table 1.
[9]See the featured map on Blockadia in the EJAtlas database: https://ejatlas.org/featured/blockadia
[10]See https://globalclimatestrike.net

investment projects grabbing their commons in many locations. It is important to emphasize that fisher people are therefore not secondary actors who join other resisting social actors demanding environmental justice in only a limited number of cases, but rather they constitute an organized social group defending their own fishing rights and seeking 'fisheries [and blue] justice' (Mills 2018).

Material and biophysical dimension of fisher conflicts

The material and biophysical dimension of fisher folk conflicts is predominantly linked to depleting fish stocks and rising marine social metabolism. The injustices in this dimension constitute significant challenges for fisher communities, usually due to four main causal factors: (i) industrial fishing practices, (ii) industrial pollution of seas and oceans, including by all sectors but especially oil spills and coastal industrial facilities, (iii) mining and energy projects placed on coasts, and (iv) increased marine social metabolism and waste generated by intensive marine aquaculture.

The first main reason causing conflict for small-scale fisher people are the practices of industrial fishing fleets and the way in which the global seafood sector operates. Throughout the world, the types of industrial fishing gears used by large-scale purse seiners and trawlers, as well as their territorial expansion into new marine frontiers, is threatening traditional and artisanal fishing practices (Doerr 2016; Pauly 2018).

As expected, a significant number of the fisher conflicts (13 out of 120 cases or roughly 11%, presented in the Supplementary Table 1) documented in the EJAtlas have emerged due to the unjust practices of industrial fishing, fisheries policies supporting them, and/or Illegal, Unreported, and Unregulated (IUU) fishing mostly performed by economically and politically powerful actors. This long list includes IUU fishing practices by Chinese industrial fleets in Ghana (see Supplementary Table 1, #108), illegal fishing in Bicol, Philippines (#104), overfishing and illegal fishing in Casamance, Senegal (#60), the impacts of fishing agreements with Chinese investors in Madagascar (#95) as well as activities of European fishing fleets (#28) and illegal fishing (#45) in the same country, European (industrial) fishing vessels in Western Sahara (#29), IUU fishing in Sherbro river in Sierra Leone (#46), industrial overfishing in Mauritania (#47), fishworkers' struggle in Kerala, India (#36), slave and forced labor in the seafood industry of Thailand (#86), individual tradable quotas affecting small-scale fishers in South Africa (#87), and finally, coastal fishers' resistance against government policies in favor of trawl fishing in Greece (#2).

Industrial fishing practices have proven to be one of the main conflictive and contested areas in the grabbing of SSF communities' resources (DuBois and Zografos 2012; Pauly 2018). The inequalities generated by this sector are often neglected or even politically encouraged by national and regional authorities, many of whom tend to assume this type of grabbing is an inherent aspect of the global seafood production system in order to provide seafood for a growing global demand. Such injustices are in fact observed and recognized by the FAO Guidelines – for securing sustainable small-scale fisheries – (FAO 2015), but not yet overcome – due to the voluntary nature of the guidelines. As discussed in the WFFP's 7th General Assembly, several fisher organizations consider the lack of enforcement and implementation of FAO Guidelines by many Member States as hinting at the absence of real political will or dedication to improving small-scale fishers' circumstances. Real and effective change would require the removal of

political and economic barriers to their subsistence and limiting the material injustices created by industrial fishing.

In most of these conflicts linked to industrial fishing, investment projects are realized through bilateral agreements between states (see the Chinese fishing fleet in Madagascar or European fishing fleet in Madagascar or Western Sahara, #28 and #29) or directly between host states and representatives of big seafood industries. Most of these agreements contain a certain degree of corruption and/or a lack of transparency about the beneficiaries of these processes (CFFA 2020), contributing to poverty and hunger on the ground. The worsening conditions of global fish stocks and increased marine social metabolism have created hostile conditions for fisher people. These conditions, however, are amplified by political transformations, such as structural adjustment programs imposed by the IMF which has led to increased poverty in Senegal, where a strong reliance on small-scale fishing and agriculture has been prevalent for ages (Vives 2017). Therefore, there are deep structural injustices – stemming from both older colonialist and newer capitalist relations – at the roots of these conflicts.

Second, industrial pollution of seas and oceans and increasing sedimentation severely affect local small-scale fishing, which is directly related to increasing marine social metabolism. Cases of fisher folk conflicts due to pollution include offshore oil spills, such as the Iranian tanker disaster and oil spill in the East China Sea (#49); the oil spill due to the collision of two tankers in the Bohai Bay near Tianjin, China (#94) – where a subsequent lawsuit ensured compensation to fisher people and became a benchmark for similar cases in China; the Taean National Park oil spill in South Korea (#93) where 70% of fisher people were reported to have to quit fishing for three years after the accident; the struggle of small-scale fisher people against contaminating industries in Valparaiso, Chile, which resulted in the assassination of fisher union leader Alejandro Castro (#88); oil extraction in the Jubilee Field in Ghana (#52); waste dumping conflicts in Bar Elias, Lebanon (#6); and in central Vietnam due to the operations of Taiwanese company Formosa Plastics Group's steel plant waste dumping into the open ocean (#105). In some instances, these conflicts emerge after the impacts of pollution have been felt and fish stocks have been devastated, whereas in other cases the potential for pollution and contamination provokes conflicts such as in Senegal, where fisher people have been opposing off-shore hydrocarbons exploitation along Grande-Côte by Kosmos Energy (#43).

The third aspect of rising marine social metabolism affecting fisher folks materializes in mining and energy projects placed on or near coasts. Mining projects uncovered in the EJAtlas adversely affecting fisher people and leading to significant conflicts are the following: black sand mining in Mac Arthur, Philippines (#56); mining and resulting contamination in Lake Titicaca, Peru (#12); seabed phosphor mining in Don Diego, Mexico, which was stopped thanks to the combined efforts of fisher people, scientists, and EJOs (#59); Takalar fisherfolk's struggle against sea-based sand mining in Sulawesi, Indonesia (#112); mining activities of Grupo Bacis in Sinaloa, Mexico, which allowed polluted water to pass through the dam known as 'El Comedero', leading to the death of fish stocks and heavy metal and arsenic contamination of community water resources, in a disaster locals call an 'ecocide' (#71); seabed manganese nodule mining in Cook Islands, New Zealand, which led to the mobilization of a wide range of grassroots groups, indigenous people, and fishers, resulting in the 'Morea Moana' ('Sacred Ocean') Act prohibiting

commercial fishing and seabed mining in protected areas close to 15 islands (#113); fisher people's mobilization with other grassroots organizations against the sand mining industry in order to preserve the Olive Ridley turtles of Kolavipalam Beach in Northern Kerala, India (#22). Energy projects also lead to unequal impacts and environmental injustices and are widely contested by fisher people, such as the Baleia wind power complex in Ceará, Brazil (#5); the proposed Nexen Aurora LNG (fracked gas) export facility on Digby Island, Canada (19); as well as other oil and gas exploration projects like in Sonda de Campeche, Mexico (#21).

Fourth, the intensive marine aquaculture is another sector increasing marine social metabolism and causing conflicts with SSF communities. The sector is usually presented as contributing to overcome overfishing crisis and supplying seafood for an increasing world population (Clausen and Clark 2005; Ertör and Ortega-Cerdà 2017, 2019). However, with its high requirements for inputs and energy, and due to its waste creation, intensive aquaculture adds to the injustices of the material and biophysical dimension. First, it increases marine social metabolism, as the biggest aquaculture companies usually focus on producing expensive carnivorous species for world markets that require high amounts of fish feed composed of high percentages of fish meal and fish oil obtained from 'cheap' small pelagic species such as anchovies or sprat. Although the industry makes efforts to reduce the fish meal and fish oil content of fish feed, for farming carnivorous species, they still need large amounts of fish content or other questionable industrial ingredients like soy or genetically modified feed (SAR 2015). Second, they need high amounts of energy for production (for a discussion on salmon farms' material and energy flows see Bogadóttir 2020). Intensive aquaculture thus serves as a conversion method of 'economically cheap' fish into a higher value commodity serving world markets, thus benefitting from an 'ecological surplus' in order to obtain higher profits (Moore 2010; Ertör and Ortega-Cerdà 2019).

A significant portion of fisher folk conflicts documented in the EJAtlas (22 out of 120) specifically result from intensive marine aquaculture, be it fish farms, shrimp production or fish feed production and the resulting pollution (see cases #3, #18, #41, #69, #80, #81, #82, and #111 from Latin America; #4, #74, #77, #78, #79, and #90 from Malaysia, India and Bangladesh; #7 and #20 from Mauritania, and #68 and #101 from Tanzania and Kenya, respectively). Other cases include regions like Greece (#58), Turkey (#97), and Canada (#92 and #37). Some of these intensive aquaculture conflicts, such as fishmeal pollution in Chimbote, Peru (#111), dirty fishmeal production in Nouadhibou, Mauritania (#20), and pollution and degradation in the RAMSAR site Kolleru Wildlife Sanctuary in India due to large scale shrimp farming for export (#4), are cases that create conflicts with fisher communities due to their impacts on rising marine social metabolism, either through increased material and energy usage or waste generation in seafood production. Other conflict cases linked to spatial justice and intensive marine aquaculture will be discussed in the next section.

Spatial justice: Ocean grabbing, displacement, and dispossession of fisher people

The second dimension of Blue Justice that I will scrutinize is *spatial justice*, which goes hand in hand with access to and control over marine commons and marine spaces.

Ocean (control) grabbing attempts through top-down investment projects and the lack of egalitarian participation mechanisms lead to restricted spaces for fishing and living. The usual result is the displacement, dislocation, and dispossession of fisher communities, creating deep injustices linked to their space, identity and attachment – pointed out as new areas of EJ scholarship and praxis by Agyeman et al. (2016). Therefore, this dimension explores both the cases in which fisher communities' access to physical space is limited or hindered and where their control over the ways and means of production, as well as their relation with the seas and oceans, are eradicated.

In this dimension, the main sectors that I will explore are intensive marine aquaculture, coastal tourism, and infrastructure projects, since they have clearer spatial characteristics when creating injustices. In most fish or shrimp farming conflicts, displacement, land dispossession, and loss of livelihoods have been reported (see #77 and #79 linked to shrimp farming in Bangladesh, #80 and # 82 in Brazil, and #18 in Ecuador). In other cases, like shrimp farming in the Muisne Estuary in Ecuador (#81), Golfo de Fonseca in Honduras (#3) and Chilika Lake in Odisha, India (#54), a loss of traditional knowledge, practices, and cultures as well as a loss of landscape and sense of place have been additional adverse socio-economic effects.

Coastal development projects leading to ocean grabbing mostly include airport and port construction, construction of artificial islands, and luxury tourism resorts on fisher peoples' and local communities' territories. New socio-technical innovations, such as an 'aerotropolis' (urbanization centered around an airport) in the Philippines (#1), deepen these inequalities and create intense conflicts.

In terms of geographical and spatial allocation, these conflicts come from a range of countries and regions. The global map demonstrates that there is a recent trend of new coastal infrastructure and other investment projects, especially in Southeast Asia, which led coastal communities and fisher people of many island states to resist massive displacements and dispossession – discussed below with concrete cases. There are also other regions (see Map 1) with an accumulation of reported conflict cases in the Eastern India with its history of long-lasting and ongoing organized fisher resistance mostly mobilized around the National Fishworkers' Forum (NFF) (Sinha 2012). In the Philippines, Indonesia, and Japan, where fishing is also an important source of identity and subsistence, fisher people are confronting expanding coastal investment projects.

The Jakarta Bay reclamation project (#107) is one such significant ocean grabbing attempt and conflict case. The project aims to construct 17 islets in the North Jakarta coastal area, which is the territory of fisher folks in Muara Baru and Muara Angke (see the EJAtlas [2019], 'Fisherfolk against reclamations in Jakarta Bay, Indonesia'). The fisher communities have felt both the ecological impacts of sedimentation and socioeconomic and political effects through a reduction of 40 to 50 percent of their incomes and the evictions of about 3,500 families in the 11 years between 2000 and 2011. One of the strongest fisher people organizations mobilized in the region, the KNTI (The Indonesian Traditional Fishermen Association), indicated that around 17,000 fishers and their families were threatened with eviction because of the project (Wardhani 2017). Due to this long-lasting mobilization, the Ministry for Maritime Affairs responded to the concerns of fisher communities and declared that fisher people would be provided with flats and boats in the islets to ensure their activities and livelihoods. However, the organization Solidaritas

Perempuan (Women's Solidarity) indicated that these compensations excluded fisherwomen who were legally not recognized as fishers but only part of fisher families (ibid.).

The Save Jakarta Bay Coalition, together with its participants such as KNTI, KIARA (People's Coalition for Fisheries Justice), and Solidaritas Perempuan, strive to give opponents of the project a strong voice in order to obtain compensation, postpone and/or cancel parts of the project. Additionally, they seek to make fishers' rights and livelihoods more visible to local authorities and foreign funders such as the Dutch embassy, due to the participation of Dutch companies in the project. Protests with slogans like '#JakartaTolakReklamasi' (Jakarta opposes reclamation) and 'Stop Reklamasi Berselimut Korupsi' (Stop corruption-tainted reclamation) became inflection points, and judicial activism resulted in some positive outcomes for fisher people, such as the cancelation of the reclamation permit for three islets. However, most of the procedural steps, including the issuance of a Building Construction Permit in 2019 for 'Islets D', have shown that fisher folks are usually marginalized and excluded from the decision-making process (Batubara, Kooy, and Zwarteveen 2018; EJAtlas #107).

Other striking mega projects documented in the EJAtlas that threaten fisher communities with spatial injustices include the construction of a deep-sea port and airport on Kalagote Island in Myanmar (#103); the Palm Islands (Palm Jumeira) in the United Arab Emirates (#62); the Colombo Port City in Sri Lanka (#33); coastal development and land reclamation in Bahrain (#9); and the Bulacan Aerotropolis in the Philippines (#1). The latter conflict, the 'aerotropolis', is a proposal for a massive airport and airport city complex whose construction threatens the livelihoods of more than 20,000 fishers in and around Bulacan and has already led to the displacement of at least 700 fisher families. These communities collaborated closely with the national fisher folk alliance Pamalakaya, universities, churches, and environmentalist groups, claiming that this project would cause massive environmental destruction (EJAtlas 2018c).

Such mega-projects usually lead to 'mega-inequalities' and deep injustices. These investment projects appropriate previously common spaces, both on the seas and land, and result in the displacement and dispossession of coastal communities, most of whom are engaged in fishing activities. Typically, fisher communities are not recognized and their participation is not ensured them being the most relevant social actors for decisions and plans regarding coastal infrastructures and the use of coastal territorial spaces. Instead, such investment projects are supported and promoted by the elites of host states, who usually establish suitable policies and political structures for profitable investments and the displacement of fisher people and local communities.

Autonomy and sovereignty of fisher people

Finally, the third dimension of Blue Justice for fisher folks is linked to *autonomy*, *self-determination*, and *sovereignty*. As argued by Martínez-Torres and Rosset (2010, 155), 'constructing new collective identities and unity along the axis of the peoples' right to self-determination' strengthens the mobilizations and political capacity of marginalized people. Fisher people and their organizations claim their right to be recognized as relevant actors in seafood production as well as to have the capability of deciding and organizing their own coastal livelihoods and maintaining their fisher identities. This dimension connects fisher struggles to 'food sovereignty' movements and illustrates their common

causes and possible (and already existing) allies (Levkoe, Lowitt, and Nelson 2017). Establishing a common identity, in this case the fisher identity, as well as defining common causes, interests, and enemies, empowers movements and allows them to construct 'counter-hegemonic projects' – as discussed by Karriem (2009) regarding the evolution of the Brazilian landless movement (MST). In this section, I will elaborate on the dimension of *autonomy and sovereignty*, first through examples of self-determination of fisherwomen's and indigenous fishers' mobilizations, then focusing specifically on alliances and movement-building.

The first important factor to be scrutinized here is the construction of autonomy and self-determination of different groups within fisher movements, helping to create horizontal structures that minimize internal power asymmetries. The importance of 'horizontal, participatory organizing approaches […] to build a transformative movement that seeds to transcend hierarchical power relations' (Karriem 2009, 323) has been discussed in the literature regarding the evolution of different social movements. For global fisher movements as well, especially the identity of fisherwomen and indigenous fisher peoples, the recognition of their rights, their sovereignty in terms of self-organization, the ability to raise their own voices, reclaiming their own needs and rights, and constructing their political agencies forms an essential political (justice) dimension.

As in the case for the internal organization of La Vía Campesina (Martínez-Torres and Rosset 2010), presumably the closest ally of global fisher movements, WFFP aims to establish similar horizontal structures and political spaces in its general assemblies with women's, indigenous, and youth sub-assemblies. This organizational structure within the WFFP aims to provide forums for the afore-mentioned fisher peoples' intersectional groups to discuss, to identify and raise their own concerns and incorporate them into the general assembly's resolutions (author's own participant observation). This is key to consolidating strong movements and networks with an emphasis on more horizontal structures and less hierarchical relations. Furthermore, to confront such injustices and ensure different groups' autonomies in the movement, a better understanding of both intersectional inequalities and injustices (Malin and Ryder 2018; Pellow 2018) and corresponding political agency and mobilization is crucial, which will be further discussed below.

Fisherwomen's resistance and mobilization play a critical role in this regard, as discussed in two recent reports by the TNI (2020) and FAO (2018). The gender inequalities prevalent in society and in most economic sectors can become strikingly deeper and critical to the livelihoods of highly marginalized groups. Fisherwomen going to capture fish or other aquatic animals with their boats and working in different phases of seafood production usually earn lower wages, receive less recognition, less social and economic protection, and are obliged to accept precarious and 'invisible' jobs – including care work – despite their formal, informal, and nonpaid labor being crucial for these sectors. The socio-economic discrimination of fisherwomen first emerges in their lower wages, where: 'women are paid (at least) 12% less than men for what appears to be the same work' (EJAtlas, #34).

However, official numbers usually do not reveal the true extent of this problem, since most women work informally or perform non-paid (and care) work (TNI 2020). Therefore, the few existing studies focusing on fisherwomen and gender inequalities in the fishing sector and available official data do not demonstrate the multidimensional work – nor the

inequalities embedded in them – undertaken by women in fishing and aquaculture sectors (Biswas 2011). They are one of the main social groups whose voices are usually silenced and whose labor is made invisible without recognition. Clearly, this is a crucial aspect of the environmental injustices faced by fisherwomen.

In order not to limit the analysis of gender dimension regarding fisherwomen to the gendered impacts and vulnerabilities resulting from gendered inequalities, I also focus on gendered political agency, fisherwomen's responses, resistances, and political organizations. This aims to echo their voice and make visible their mobilizations, which is still a less developed debate in peasant studies, as argued by Martínez-Torres and Rosset (2010). Fisherwomen's (*Perempuan Nelayan in Indonesian*) struggle for their recognition and rights in Indonesia as documented in the EJAtlas (#34) gives us key insights on the injustices they face and their mobilization strategies. First, they demand equal legal rights like male fishers and to be recognized as 'fisherwomen' in Indonesian law and legislation. The lack of such legal recognition has been a major source of discrimination and injustice towards them, as they were not allowed to apply for government subsidies nor for insurance. A feminist analysis of Indonesian Law on the protection and empowerment of fisher*men* (emphasis added) and the regulations of the Ministry of Marine Affairs and Fisheries highlights the struggle of two fisherwomen and how they obtained a legal identity – and thus recognition and legitimacy as fisherwomen – through their fisher cards and insurance (Ismail 2017).

Second, through their network, the PPNI (the Women's Fisheries Sisterhood of Indonesia) connects 16 fisherwomen groups in Indonesia. They also form part of the People's Coalition for Fisheries Justice (KIARA)[11] in Indonesia, a non-governmental organization established in 2003, which advocates for SSF communities' rights. The research conducted by the KIARA and PPNI uncovered that in Indonesia more than 56 million people are involved in fishing activities, including catching, processing, transporting, and marketing. Women account for 70% (about 39 million people) of these fisher people with most working more than 17 hours per day, while their contribution to family incomes reaches 48 percent (KIARA 2015; EJAtlas #34[12]).

As a network, the PPNI and its women members have also been participating in struggles affecting SSF people, such as against the Jakarta Bay Reclamation Project, in order to have their own voice and make it known that these investment projects have substantial negative effects on women (ibid.). Moreover, they also participate in the General Assemblies of WFFP and participate directly in the global fisher folk movement (TNI 2020).

Indigenous people form another important social actor in the *autonomy* and *self-determination* of fisher people. Four cases stand out in the EJAtlas in which indigenous fisher people have led the struggle. Two of these are mobilizations in Canada, providing important insights about indigenous fisher struggles, where the Grassy Narrows and Whitedog First Nations oppose mercury poisoning (#40) and the Listuguj First Nation defends their fishing and logging rights (#109). In the third case (#117), the Sonaha indigenous minority groups in Southwestern Nepal struggle for their traditional livelihoods and fishing that

[11]See the webpage of KIARA: http://www.kiara.or.id; and a media article on their movement: Guerrero and Herawati (2017).
[12]See information on this conflict in the EJAtlas (2018a).

had been crippled with the creation of the Bardia National Park and the intervention of armed park rangers. The park was created in 1976 and displaced many landless Sonaha from their customary territories, forcing them to become wage laborers and seasonal agricultural workers. Since the late 1990s, their mobilization became stronger and they engaged in alliances with NGOs. A resulting wider campaign between 2006 and 2011 allowed them the right to fishing licenses for nine months. However, the conflicts have continued and the so-called participatory mechanisms have not led to their full practice of traditional fishing and real recognition of their territorial rights, continuing a trend of labor outmigration. Finally, the Itelmen of Kovran on the Kamchatka peninsula of Russia, fight for their fishing rights which are being undermined by commercial fishing companies (#96).

These are not the only conflicts where indigenous people make claims for their fishing rights. Rather, in fisher movements such as the WFF and WFFP, they have their own say, and depending on the territory, they are the ones engaging in traditional fishing practices. However, their situation requires a close look at how intersectionality plays a role in marginalized communities and their struggles, as pointed out by other environmental justice (Malin and Ryder 2018) and critical environmental justice scholars (Pellow 2018) who intend to highlight structural inequalities. As studies on the evolution and transformation of the La Vía Campesina (Martínez-Torres and Rosset 2010) and MST (Karriem 2009) point out, the alliances between peasant (and fisher) movements and indigenous movements are of the utmost importance, especially for scaling up and strengthening their struggles.

Alliances and movement-building

Alliances to confront uneven transformations of both rural and urban spaces, as well as environmental injustices in different contexts, form the basis of strong social movements claiming social and environmental justice and proposing alternative social and economic models (Karriem 2009; Batubara, Kooy, and Zwarteveen 2018). In his analysis of the Brazilian Landless Workers' Movement (MST), Karriem (2009, 324) highlights the importance of 'alliances across interrelated scales to unite a diverse range of social groupings' in order to build a counter-hegemonic project. In line with this approach, the three dimensions of Blue Justice proposed in this framework aim to both understand the underlying reasons of such environmental conflicts as well as to highlight the ways in which they are contested, where alliance building weaves strong links for empowering marginalized actors and for emancipatory politics.

One such striking case of *alliance and movement building* on a regional scale is illustrated by the fisher movement in Kerala, India (#36). Both Kerala's growing marine social metabolism in order to serve global markets and fisher mobilizations go back to the 1960s, when the Indian government undertook measures to encourage export-oriented prawn fisheries, mostly with industrial trawlers. This shift transformed fisheries in Kerala from traditional small-scale enterprises to an industrial company-driven model, which marginalized coastal fisher communities. Similar to the evolution of the MST in Brazil (Karriem 2009), there was an initial period in the 1970s where religious leaders were at the forefront of mobilization. Following that, in the 1980s the fisher movement became more autonomous from political parties and religious leaders and changed its name to 'Kerala Swathantra Malsya Thozhilali Federation' (The Kerala Independent

Fishworkers' Federation), adopting the term 'fishworkers' (malsya thozhilali) and converting itself to a secular trade union.[13] This expanded its base of social actors and made links with worker movements on the rural-urban connection (for its detailed analysis see Sinha 2012).

The period was cauterized by intense conflict, with trawlers entering SSF's fishing territories and aggressively attacking them, resulting in the murder of one fisherman, Babu, after an industrial boat hit his small boat. The struggle continued into the 1980s and onwards, with numerous hunger strikes, protests, arrests, marches, and agitations, with demands issued to political authorities such as the Ministry of Fisheries and the Ministry of Finance, including compensation for Babu's family (EJAtlas, #36). These continuous mobilizations resulted in two crucial legal victories in the 1990s. The first was a ban on trawl fishing during the monsoon season between the 20th of July and 31st of August in the territorial waters of Kerala. The second was a decision issued by the Supreme Court in 1993, which reaffirmed the trawling ban imposed by the government during the monsoon season. Further political engagements and alliances were built by the Kerala Independent Fishworkers' Federation and National Fishworkers Forum (NFF) in mobilizations against free trade agreements, participation in anti-nuclear movements, and the formation of National Alliance of People's Movements in India and the International Collective for the Support of Fishworkers (EJAtlas, #36). These cases exemplify the material, political, and legal injustices faced by many traditional fishing communities.

As a result of the analysis of fisher conflicts from the perspective of the three Blue Justice dimensions, I argue that fisher people are mobilized and relevant actors resisting (ocean grabbing/investment) projects on the ground and form part of the 'global environmental justice movement' as defined by Martinez-Alier et al. (2016). Fisher movements demonstrate their blue layers and strive for fisheries justice, which is currently strongly linked to the food sovereignty movement (for a discussion on interlinked alliances of environmental justice movements, see Agyeman et al. 2016; for fisheries and climate justice, see Mills 2018). The mobilizations and claims of environmental justice (including climate justice, fisheries justice, and food justice) at different scales easily echo the claims of the food (as well as energy) sovereignty movements. Fisher people, with their varying organizations, have already become part of food sovereignty movement(s) and allies of mobilized peasants, pastoralists, and beekeepers, among others. Considering the scale and interrelatedness of these struggles**,** food sovereignty offers a key umbrella concept and the possibility of bringing together both rural and urban actors – without intervening in or occupying each other's political spaces. Scaling up these movements together with urban social actors, including consumer cooperatives, EJOs, academics, and activists, is on the agenda of such global movements (TNI 2020), and fisher movements play their role in such an endeavor.

Concluding remarks

This article has provided a global overview and analysis of 120 conflicts faced by small-scale fisher communities. In order to understand better their local and global struggles

[13]See information on this fisher struggle in the EJAtlas (2018d).

and their political significance, I provided a Blue Justice framework based on three dimensions: *(i) material and biophysical dimension, (ii) spatial justice,* and *(iii) autonomy and sovereignty*. This theoretical framework aims to contribute to and expand environmental justice practice and scholarship by enhancing our understanding of the mobilizations of an understudied and marginalized social actor: small-scale fishers and their communities. It first enables discovery of the underlying material and structural causes of these conflicts and their links to the politics of the current food production system. Secondly, it provides a conceptual lens to unfold the political agency of fisher people and their potential of scaling up their movements and gaining further autonomy and sovereignty through their already existing alliances.

My main argument in this article is that fisher people engage in substantial organized resistance as mobilized social actors opposing a range of investment projects and (fisheries and) environmental injustices on the ground as well as striving for their rights for fishing, their fisher identity and territorial and communal rights. Although they are utterly marginalized and made almost invisible by national, regional, and global policies in many places, they continue to organize, form alliances, and are able to stop part of the projects they fight against.

As such, in this article, I aimed at uncovering the blue layers of environmental justice, demonstrating their links as struggles against ocean and land grabbing attempts, and highlighting the common paths and goals of justice and sovereignty movements (especially around primary food production but also increasingly linked to urban movements). This article thus intends to contribute to the discussions on the environmental justice and sovereignty struggles towards a more radical political transformation of society as a whole.

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